Natural Biogeochemical Cycles and the influence of anthropogenic perturbations on key elemental cycles

by
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Why study biogeochemical elemental cycles?

- Utmost importance for the sustenance and balance of ecosystems and the provision of ecosystem services
- Regulate the availability and distribution of essential elements required for life
- Impacts climate and weather patterns
- Understanding how ecosystems respond to anthropogenic activities
- Contribute to the regulation of nutrient cycles in agricultural systems, impacting food production and global food security

Introduction

- Cycling of elements on Earth involves

  - **Geological and geochemical processes**, such as erosion, water drainage/runoff, movement of the continental plates & weathering…

  - **Chemical and biological processes**

  - **Interactions between biotic and abiotic components**

- Chemical elements move through both biotic (i.e., biosphere) and abiotic compartments (atmosphere, hydrosphere, and lithosphere) on Earth - Inputs, outputs, sources, and sinks

- Elements (such as C, N, P, S, etc.) take a variety of chemical forms, and are cycled and reused within and among Earth's various compartments over and over again – (natural) **Biogeochemical cycles.**

- Along with energy flows, biogeochemical cycles establish the relations among ecosystem compartments (i.e., biotic and abiotic) at local, regional, and global scales.

- The total biomass, metabolic diversity, and presence in almost all habitats on earth signify the crucial and dominating roles of microbes in element cycling.

The Global Carbon Cycle

- Carbon reservoirs on Earth

- Carbon cycling within and between reservoirs - Key processes

- Anthropogenic effects on the global carbon cycle/budget

CO_2, a fully oxidized version of carbon serves as the basic building block

Major ‘C’ reservoirs

Major Carbon Reservoirs on Earth

Reservoir	Percent of Total[a]
Rocks and sediments	99.5[b]
Oceans	0.05
Methane hydrates	0.014
Fossil fuels	0.006
Terrestrial biosphere	0.003
Aquatic biosphere	0.000002

[a]Total carbon, 76×10^{15} tons
[b]80% inorganic

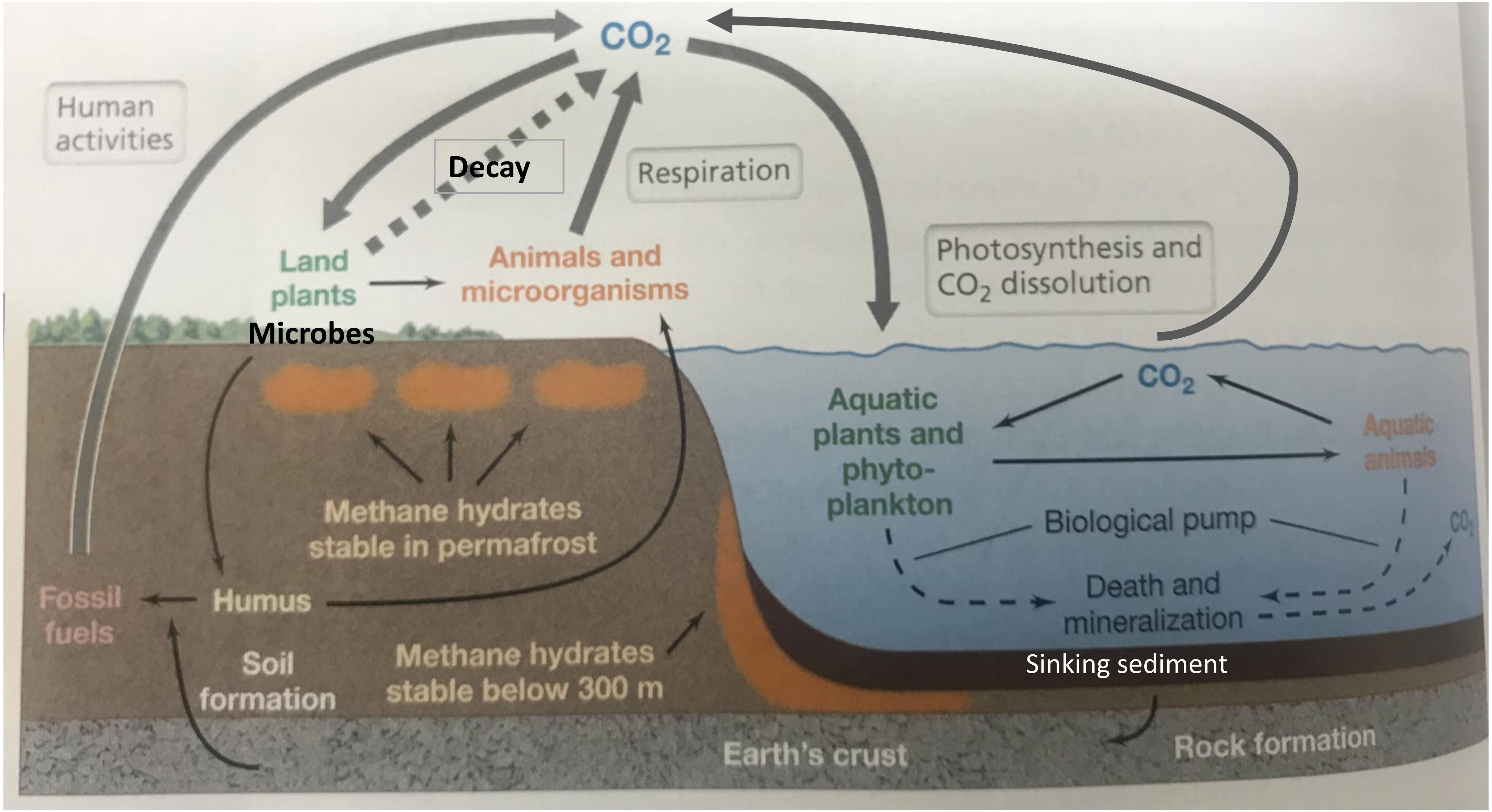
Brock Biology of Microorganisms, 15th edition

Pools	Quantity (Gt)
Atmosphere (CO$_2$, CH$_4$, CO)	720
Oceans	38,400
Total inorganic	37,400
Surface layer	670
Deep layer	36,730
Total organic	1,000
Lithosphere	
Sedimentary carbonates	>60,000,000
Kerogens	15,000,000
Terrestrial biosphere (total)	2,000
Living biomass	600–1,000
Dead biomass	1,200
Aquatic biosphere	1–2
Fossil fuels	4,130
Coal	3,510
Oil	230
Gas	140
Other (peat)	250

Falkowski et al. 2000, Science

Carbon cycle

Spanning atmosphere, biosphere, hydrosphere & geosphere



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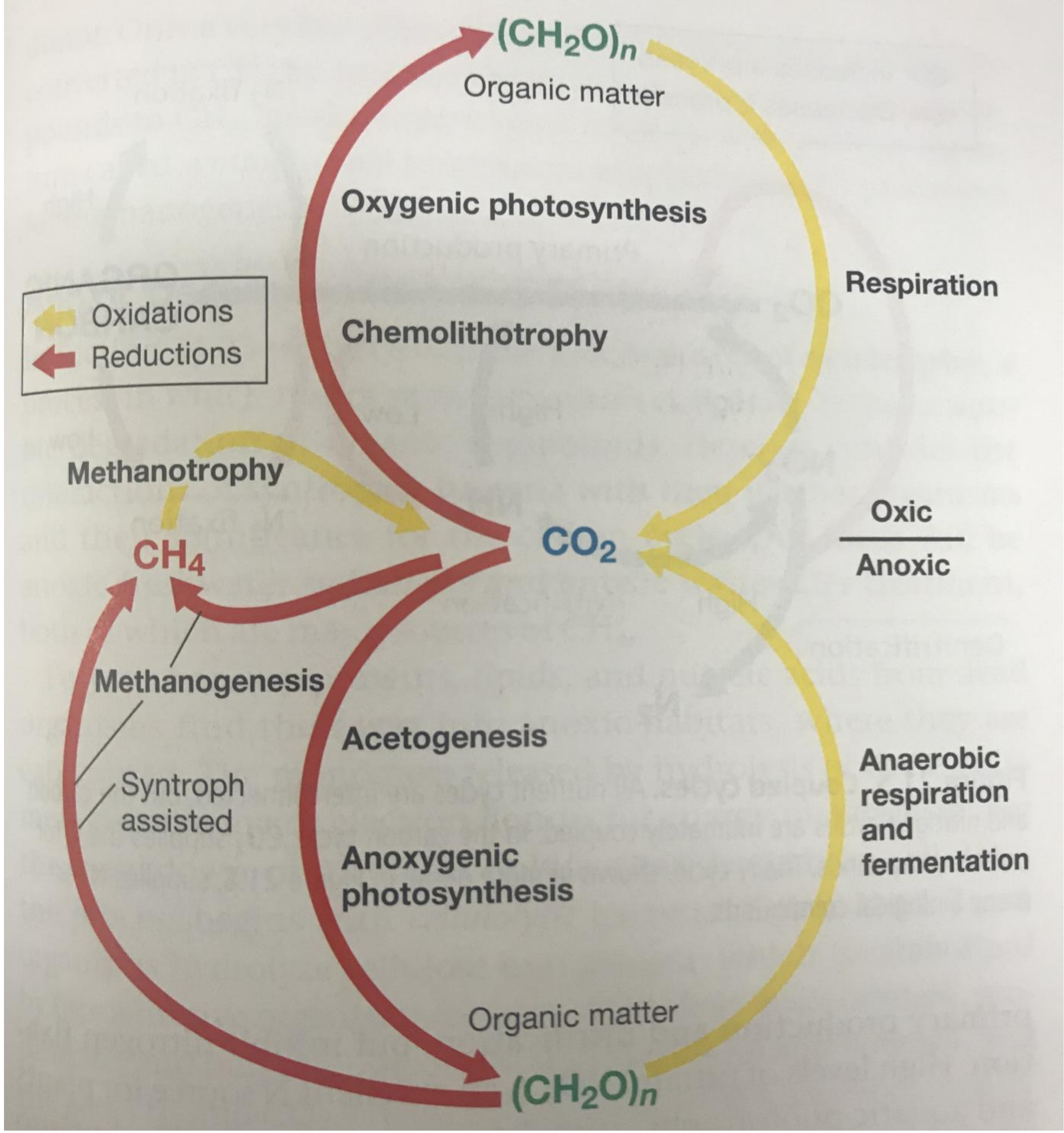


Redox cycle for carbon

It contrasts
Autotrophic
(CO_2 ----> organic compounds)
and
Heterotrophic
(organic compounds --> CO_2)
processes

$(CH_2O)_n$ represents organic matter

Brock Biology of Microorganisms, 15th edition

Key processes

- Sedimentation, volcanism, uplifting, erosion/run-off, etc. (Geological)
- Ocean exchange, Dissolution, weathering, etc. (Geo/Chemical)
- **Photosynthesis, Respiration, Decomposition, etc**. (Biological) – major role in cycling of carbon between “fast” reservoirs
- Fossil fuel burning & Land use changes (Agriculture activities, Forestation/ Deforestation, etc.) (Anthropogenic)

1 Gigatonne (Gt) = 1 billion tonnes = 1×10^{15}g = 1 Petagram (Pg)

1 kg carbon (C) = 3.664 kg carbon dioxide (CO_2)

1 GtC = 3.664 billion tonnes CO_2 = 3.664 $GtCO_2$

GLOBAL CARBON project

Fate of anthropogenic CO_2 emissions (2014–2023)

Sources = Sinks

9.7 GtC/yr
90%

10%
1.1 GtC/yr

5.2 GtC/yr
48%

29%
3.2 GtC/yr

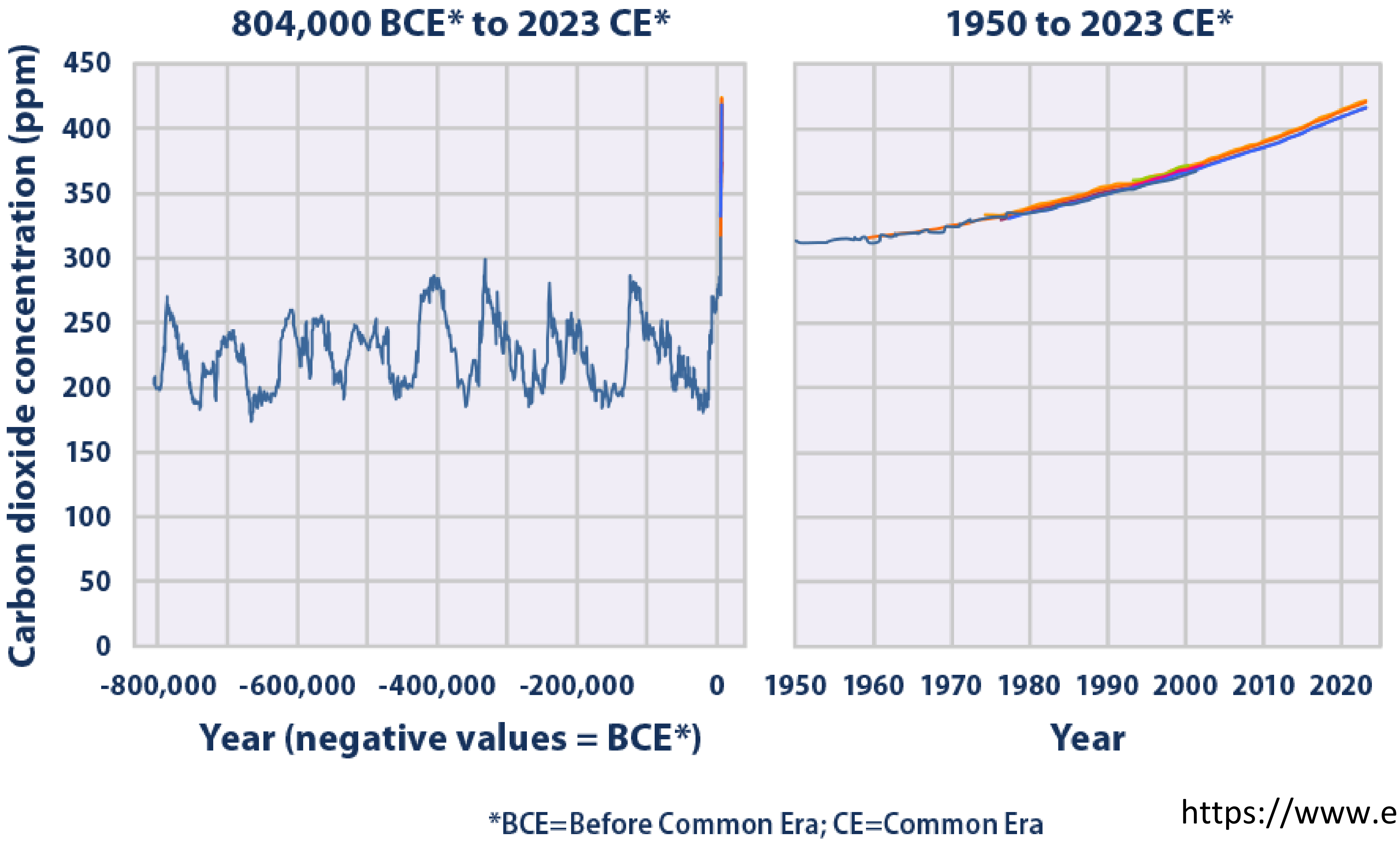
26%
2.9 GtC/yr

Budget Imbalance:
(the difference between estimated sources & sinks)

4%
-0.4 GtC/yr

Source: Friedlingstein et al 2024; Global Carbon Project 2024

Atmospheric CO_2 concentration

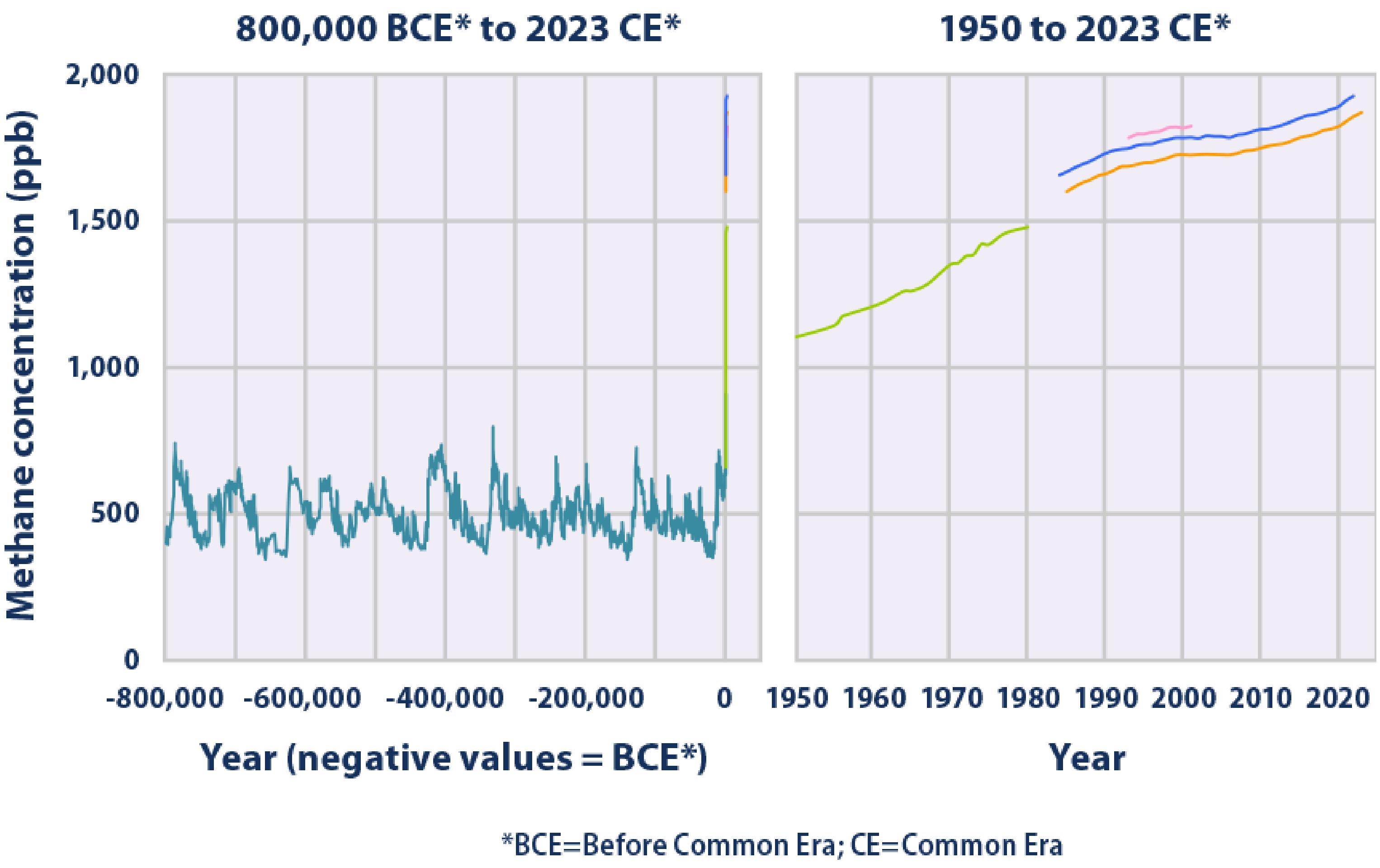


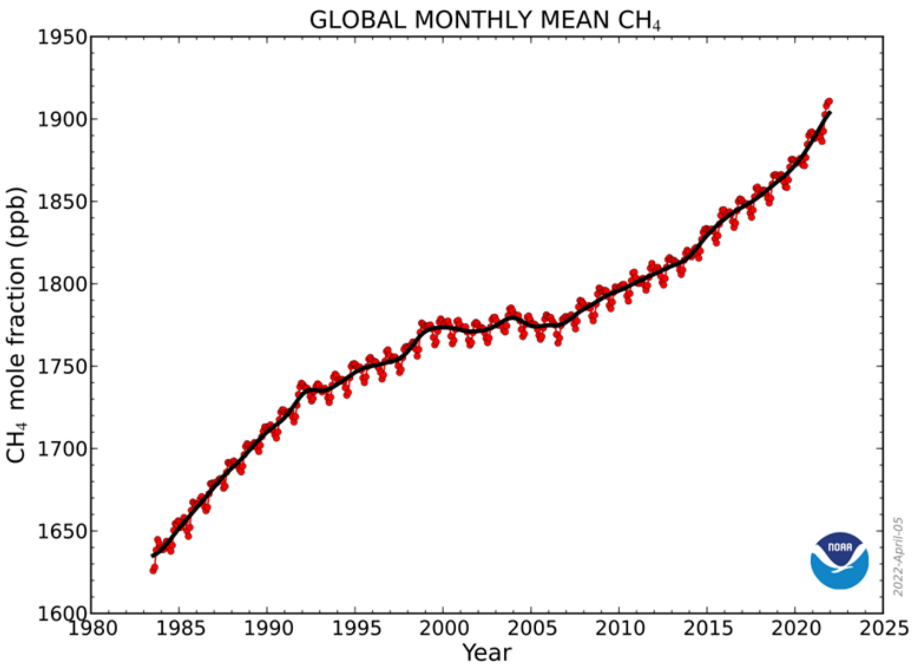
CO_2 rise from an annual average of 280 ppm in the late 1700s to >422 ppm in 2024 — a >50 % increase.

~1.8 ppmV/yr (4 PgC/yr) increase rate

https://www.epa.gov/climate-indicators

Methane – about 1% of the atmospheric carbon budget (~3 Pg C)





(NOAA Global Monitoring Laboratory)

https://www.epa.gov/climate-indicators

- **Carbon monoxide:**

CH_4 oxidation is one of the sources of CO in the atmosphere – 0.05 to 0.20 ppmv (~ 0.2 Pg C)

- Other forms (e.g., terpenes, hydrocarbons, dimethyl sulfide, etc.) – 0.05 Pg C

Oceanic carbon

- Dissolved **inorganic**, dissolved organic, particulate organic, etc.

- Governed by the Carbonate chemistry

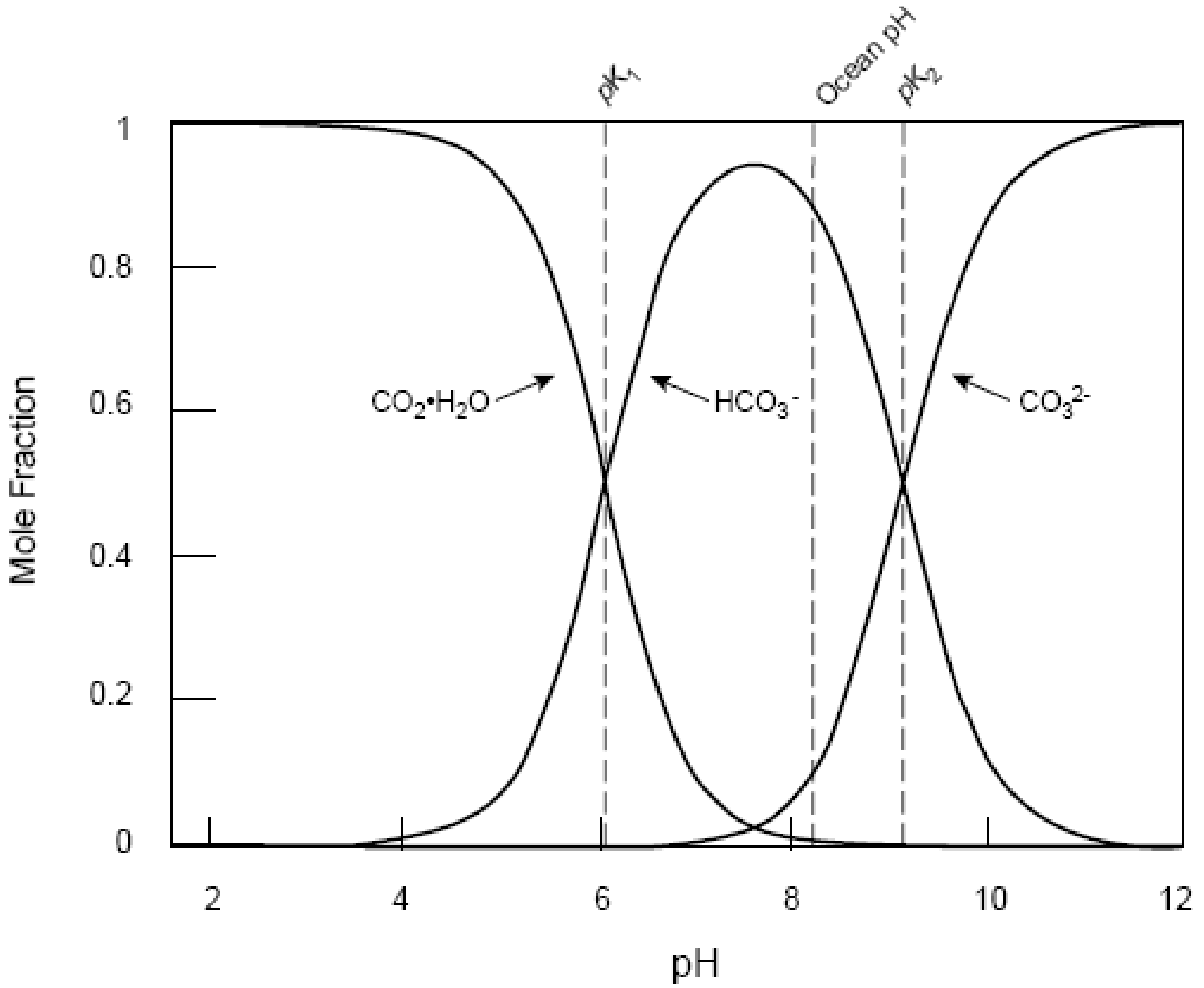
$$CO_2(g) \overset{H_2O}{\Leftrightarrow} CO_2 \cdot H_2O$$

$$CO_2 \cdot H_2O \Leftrightarrow HCO_3^- + H^+$$ **pKa = 6.1**

$$HCO_3^- \Leftrightarrow CO_3^{2-} + H^+$$ **pKa = 9.2**

- Average Ocean pH = 8.2; maintained by weathering of basic rocks

Speciation of total carbonate, CO_2 (aq) in seawater *vs* pH



Since pKa1< pH < pKa2 , most of the dissolved CO_2 exists as HCO_3^-

Ocean acidification

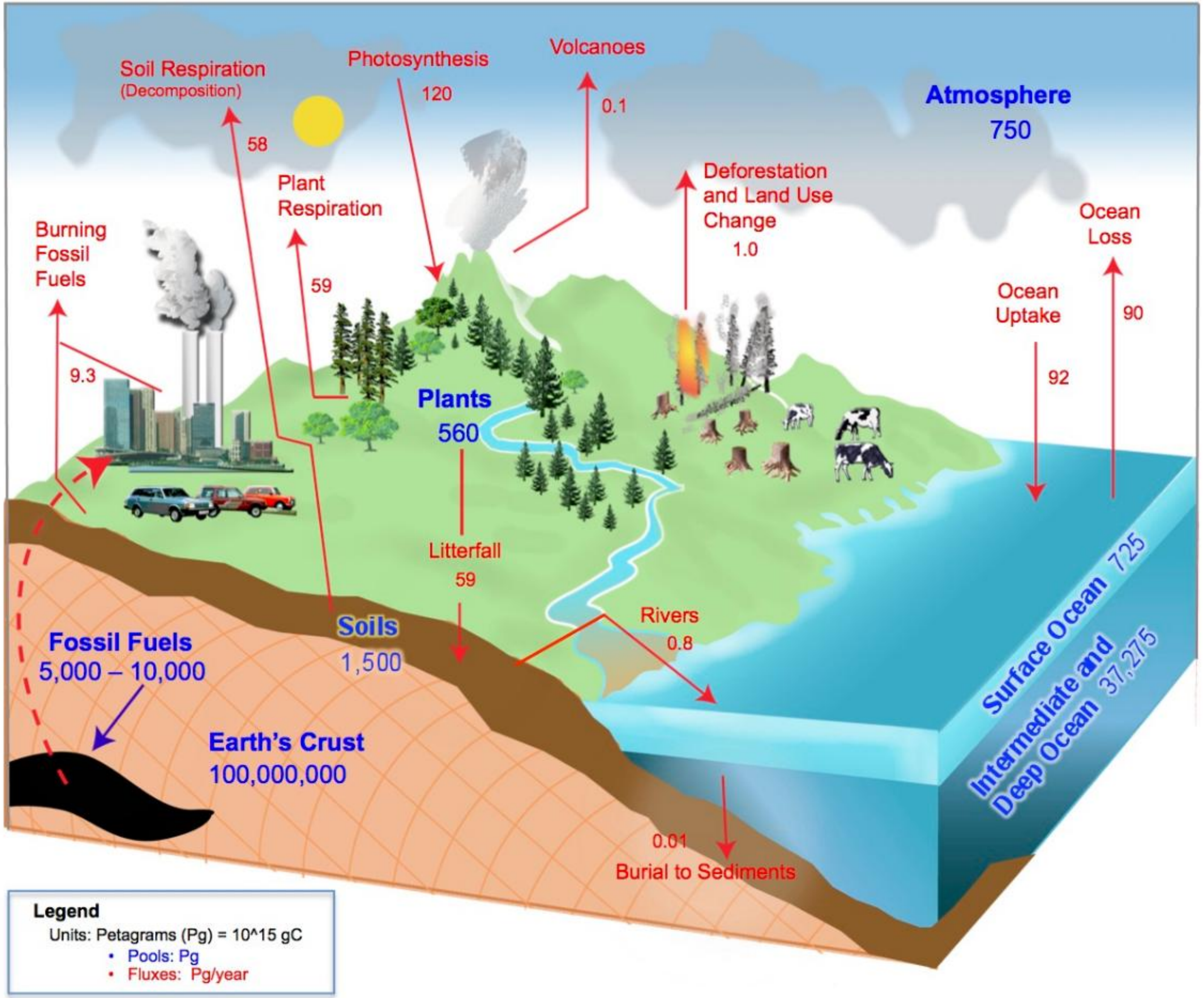
- from 8.19 to about 8.05

Note: If oceans were acidic, almost all CO_2 would partition to the atmosphere.

Global Carbon Budget:
Natural and anthropogenic fluxes

Methane flux to the atmosphere
~0.5 Pg C/yr

1 Pg C = 1 Gt C

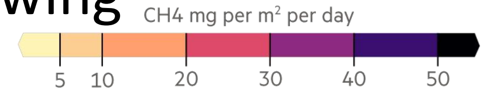


GLOBE®2017 Global Carbon Cycle Diagram Biosphere

Data Sources: Adapted from Houghton, R.A. Balancing the Global Carbon Budget. Annu. Rev. Earth Planet. Sci. 007.35:313-347, updated emissions values are from the Global Carbon Project: Carbon Budget 2017.
Diagram created by a collaboration between UNH, Charles University and the GLOBE Program.

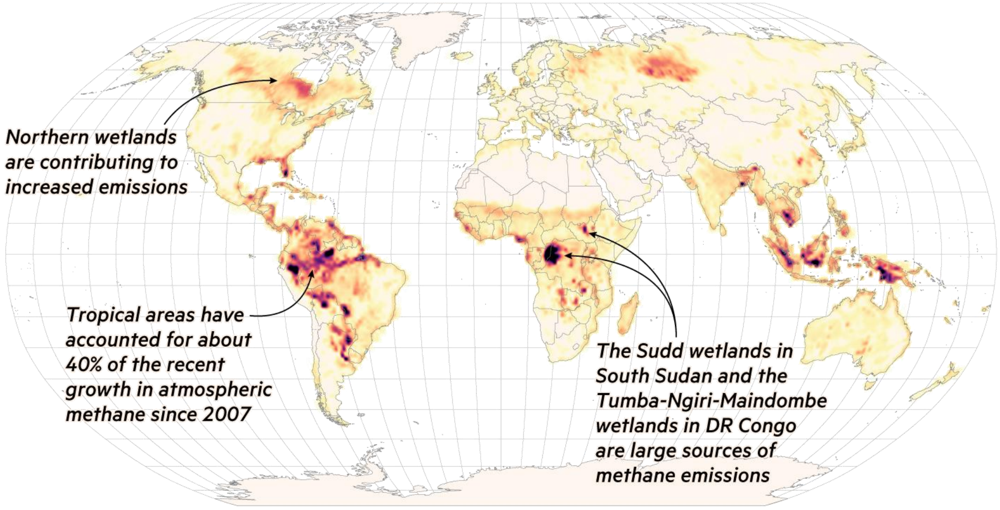
Other sources: Permafrost thawing
Methane hydrates

Methane emissions by source (2008-2017 period)



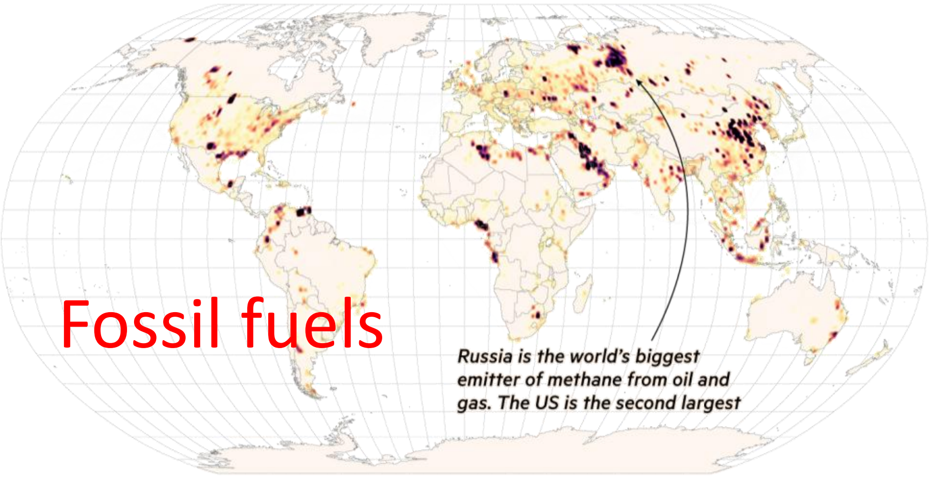
Natural wetlands

Natural wetlands



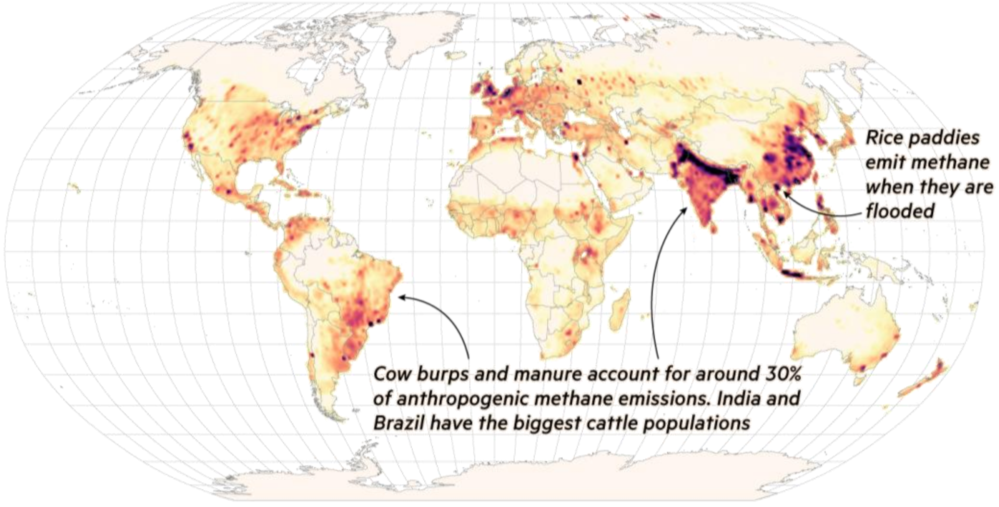
Fossil fuel

Fossil fuels



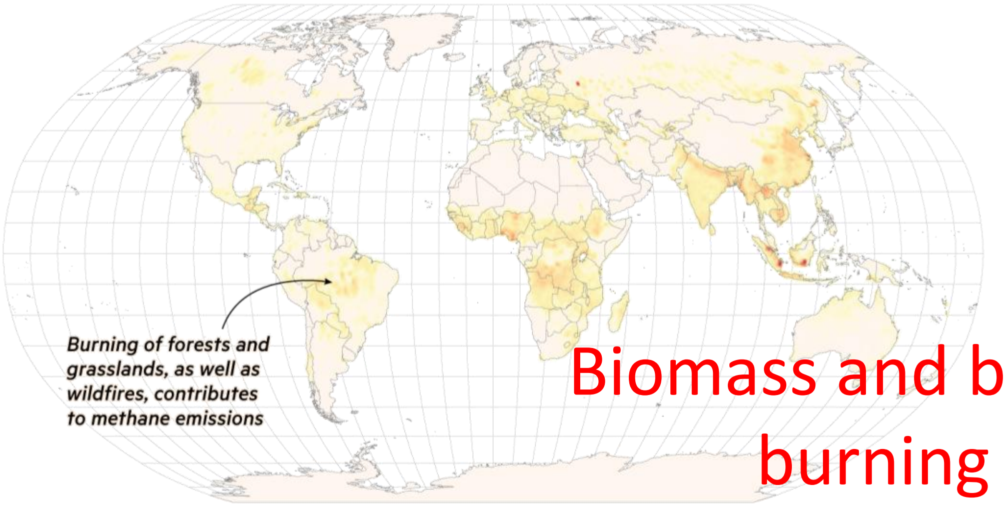
Agriculture and Waste

Agriculture and waste



Biomass and biofuel burning

Biomass and biofuel burning



Source: The Financial Times Limited 2022

https://insideclimatenews.org/

Sources: Saunois et al., Global Carbon Project; Basu et al., Estimating Emissions of Methane Consistent with Atmospheric Measurements of Methane and δ13C of Methane; International Energy Agency; FT research

Nitrogen cycle

- "N" is the most common element in Earth's atmosphere, yet it (fixed or available form) is one of the most limiting nutrients for primary production.

- It is an important component of biological macromolecules and atmospheric processes.

- Anthropogenic processes - major pollutants of water bodies and the atmosphere

- Biological transformations contribute majorly to N fluxes across major reservoirs.

Biogeochemically important N compounds and their oxidation states

Compound	Name	N oxidation state
NH_2, NH_4^+	Ammonia, ammonium	–3
R-NH_3	Amino acids, organic nitrogen polymers	–3
NH_2OH	Hydroxylamine	–1
N_2	Dinitrogen gas	0
N_2O	Nitrous oxide	+1
NO	Nitric oxide	+2
HONO, NO_2^-	Nitrous acid, nitrite	+3
NO_2	Nitrogen dioxide	+4
N_2O_5, HNO_3, NO_3^-	Dinitrogen pentoxide, nitric acid, nitrate	+5

$$\underset{[+5]}{NO_3^-} \leftrightarrow \underset{[+4]}{NO_2(g)} \leftrightarrow \underset{[+3]}{NO_2^-} \leftrightarrow \underset{[+2]}{NO(g)} \leftrightarrow \underset{[+1]}{N_2O(g)} \leftrightarrow \underset{[0]}{\boxed{N_2}} \leftrightarrow \underset{[-3]}{NH_4^+}$$

Ward B., 2012, The Global N cycle

Major Nitrogen reservoirs on Earth

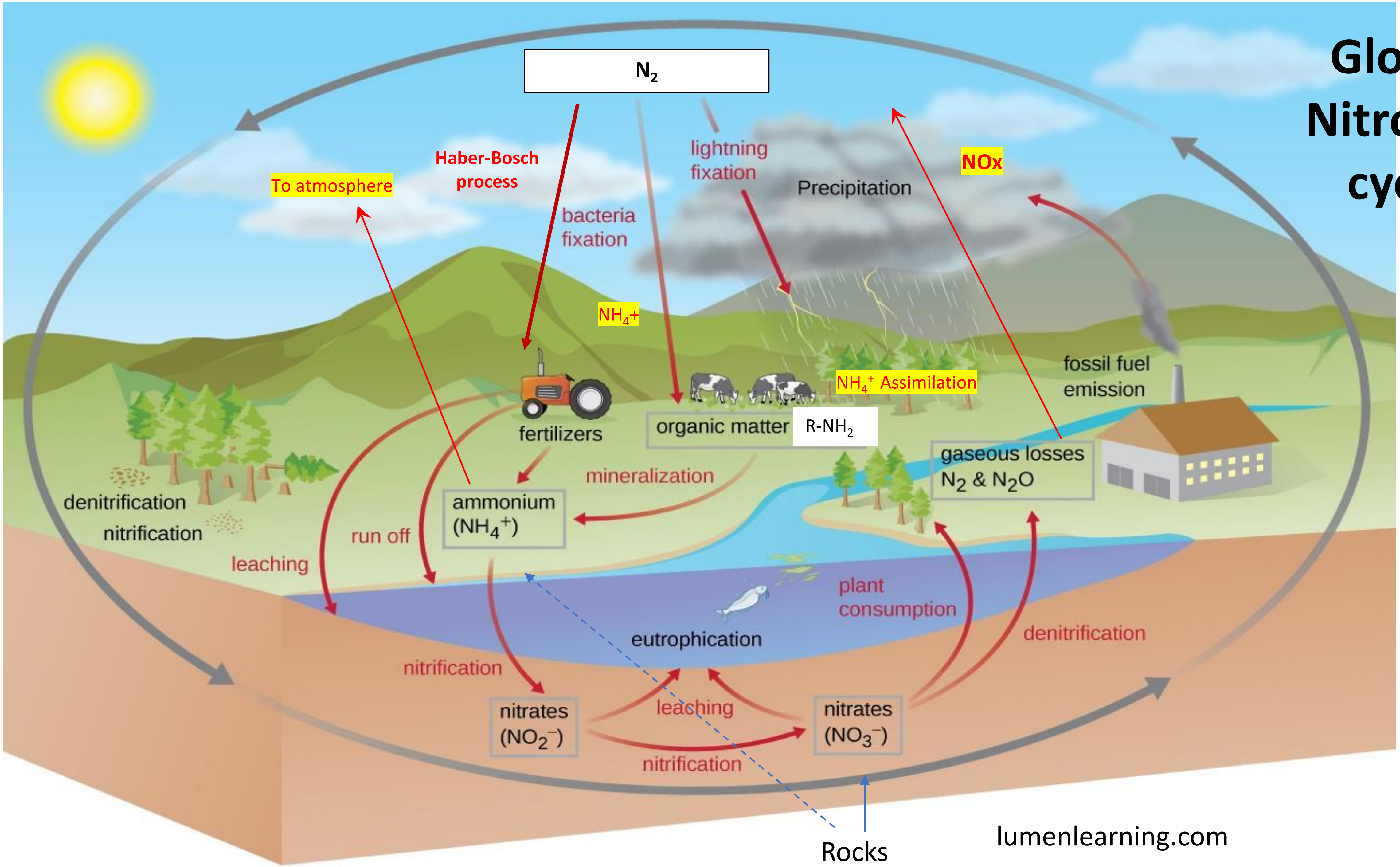
PON/DON:
Particulate/Dissolved Organic Nitrogen

*1000 Teragrams (Tg) = 1 Gt

Ward B., 2012,
The Global N cycle

Reservoir		Tg *	
Atmosphere	N_2	3.7×10^9	~ 27%
	N_2O	1.4×10^3	
Biosphere	Marine	3.0×10^3	
		5.0×10^2	~ 0.000062%
	Terrestrial	5.4×10^3	
		2.9×10^4	
		7.7×10^3	
Ocean	N_2	1.46×10^6	
	N_2O	0.34	~ 0.005%
	NO_3^-	6.0×10^5	
	PON	9.0×10^4	
	DON	8.1×10^5	
Geological	Continental crust	1.3×10^9	
	Crustal rocks	6.4×10^8	
	Oceanic crust	8.9×10^5	~ 73%
	Coastal sediments	3.2×10^4	
	Deep ocean sediments	2.0×10^9	
	Soil	1.4×10^5	
		2.2×10^4	

Global Nitrogen cycle



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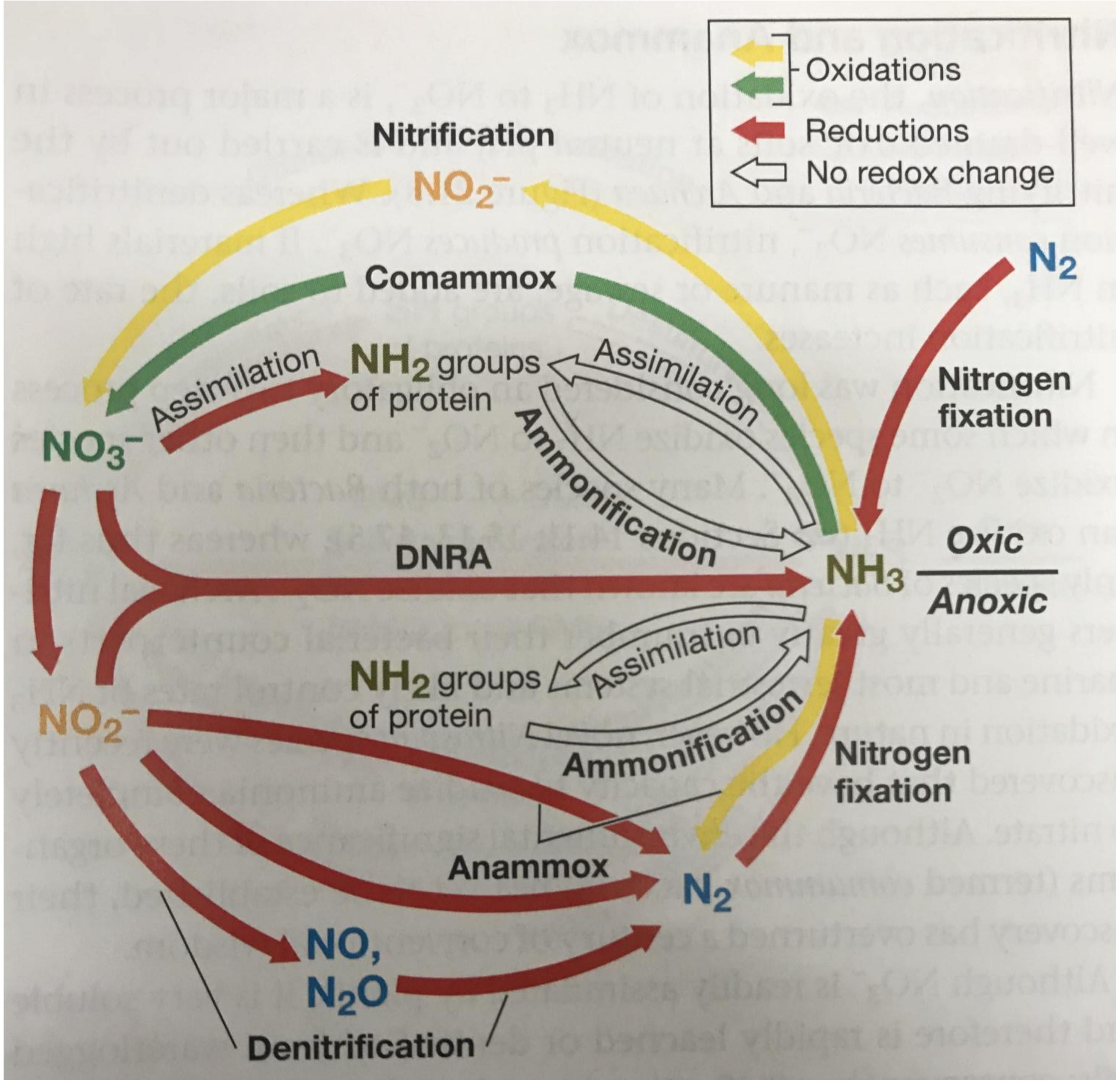


Redox cycle for nitrogen

DNRA: Dissimilatory nitrate reduction to ammonium

Anammox: Anaerobic ammonia oxidation

Comammox: Complete ammonia oxidation

Brock Biology of Microorganisms, 15th edition

Implications of denitrification

- N_2 and N_2O are released to the atmosphere – thereby completing the N cycle.
- **Detrimental process for agriculture**, as fixed N from soil gets removed (in particular in waterlogged soils) – fixed N loss
- N_2O can be photochemically oxidized to NO in the atmosphere - NO reacts with Ozone (O_3) to form NO_2^-, which returns to earth as nitric acid **(acid rain process)** – increase acidity of soils
- N_2O is a potent greenhouse gas (>270 times than CO_2 over a 100-year time frame)
- **Useful process in wastewater treatment** - to avoid N influx in water bodies and eventual eutrophication

Major processes in active N cycling

- **Biological** – several processes from N_2 fixation, assimilation, decomposition, to denitrification

- **Anthropogenic**

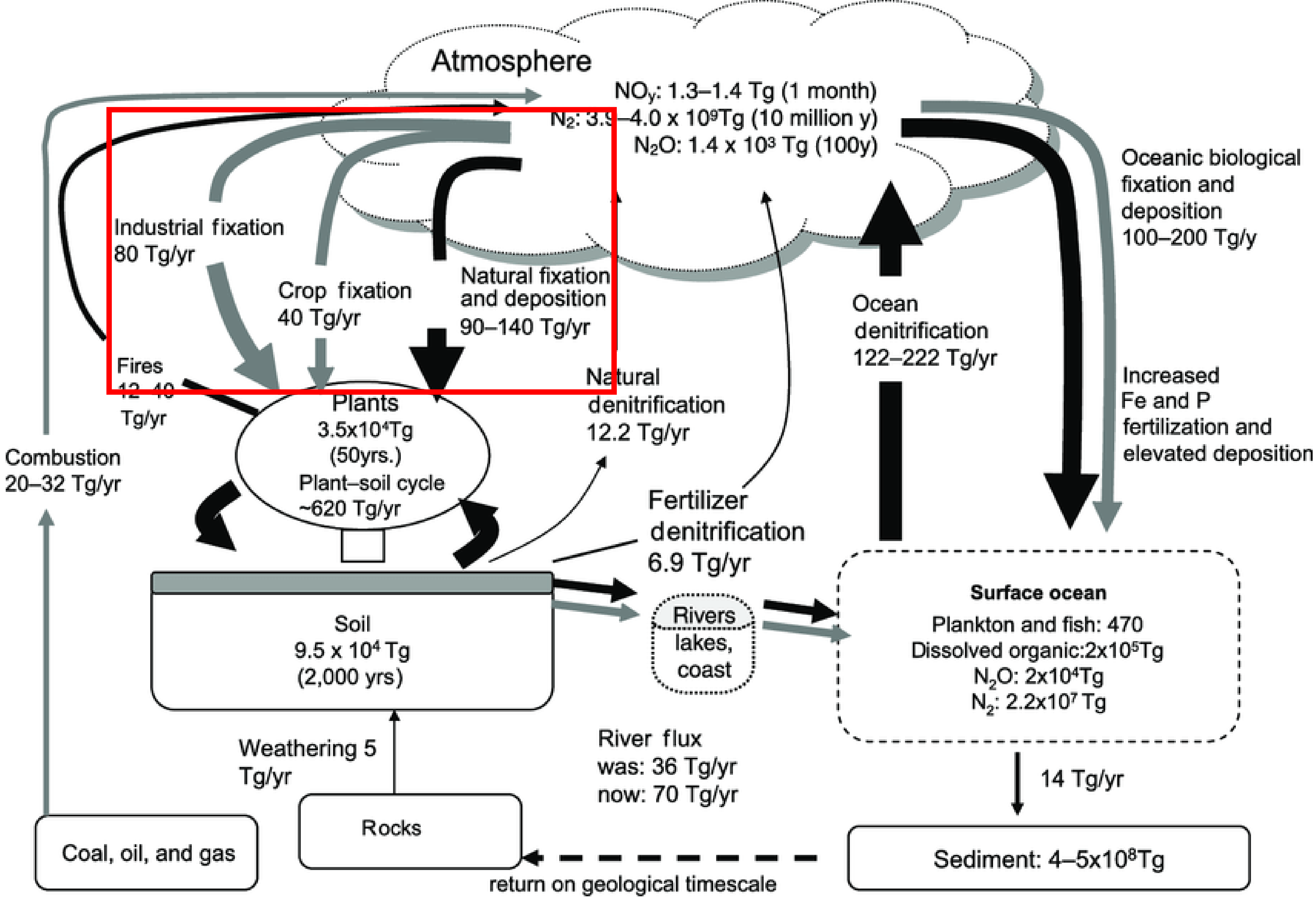
  - Haber-Bosch process; intentional production of NH_3

  - Combustion of fossil fuels/biomass; unintentional production of NOx

  - Industrial agriculture; intensification of biological N_2 fixation

- **Atmospheric**

  - several reactions; though fluxes are smaller than biological fluxes, these are important for climate, stratospheric ozone, and photochemical smog



Key Nitrogen pools (Tg), fluxes (Tg/yr) & Turnover times (y – in parentheses)

Solid arrows: Pre-industrial cycle

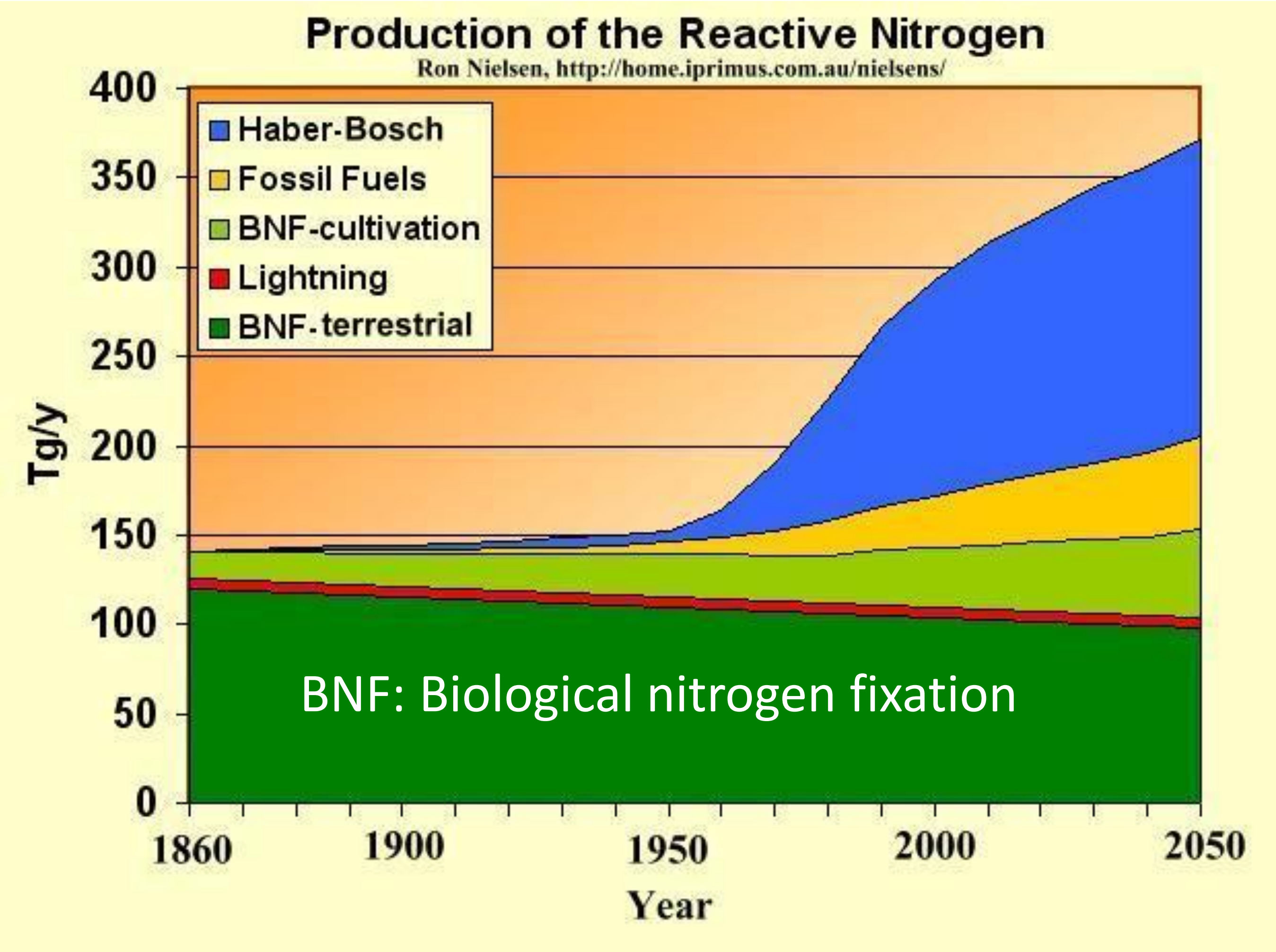
Grey arrows: Human additions

*The width of the arrows is approximately proportional to the flux rates

*To convert Tg N to moles N, multiply by 7.1×10^{10}

Source: Nutrient cycling, by A.M. Kooijman, L.B. Sparrius, and J. Sevink, In: Inland Drift Sand Landscapes, 2010, KNNV Publishing

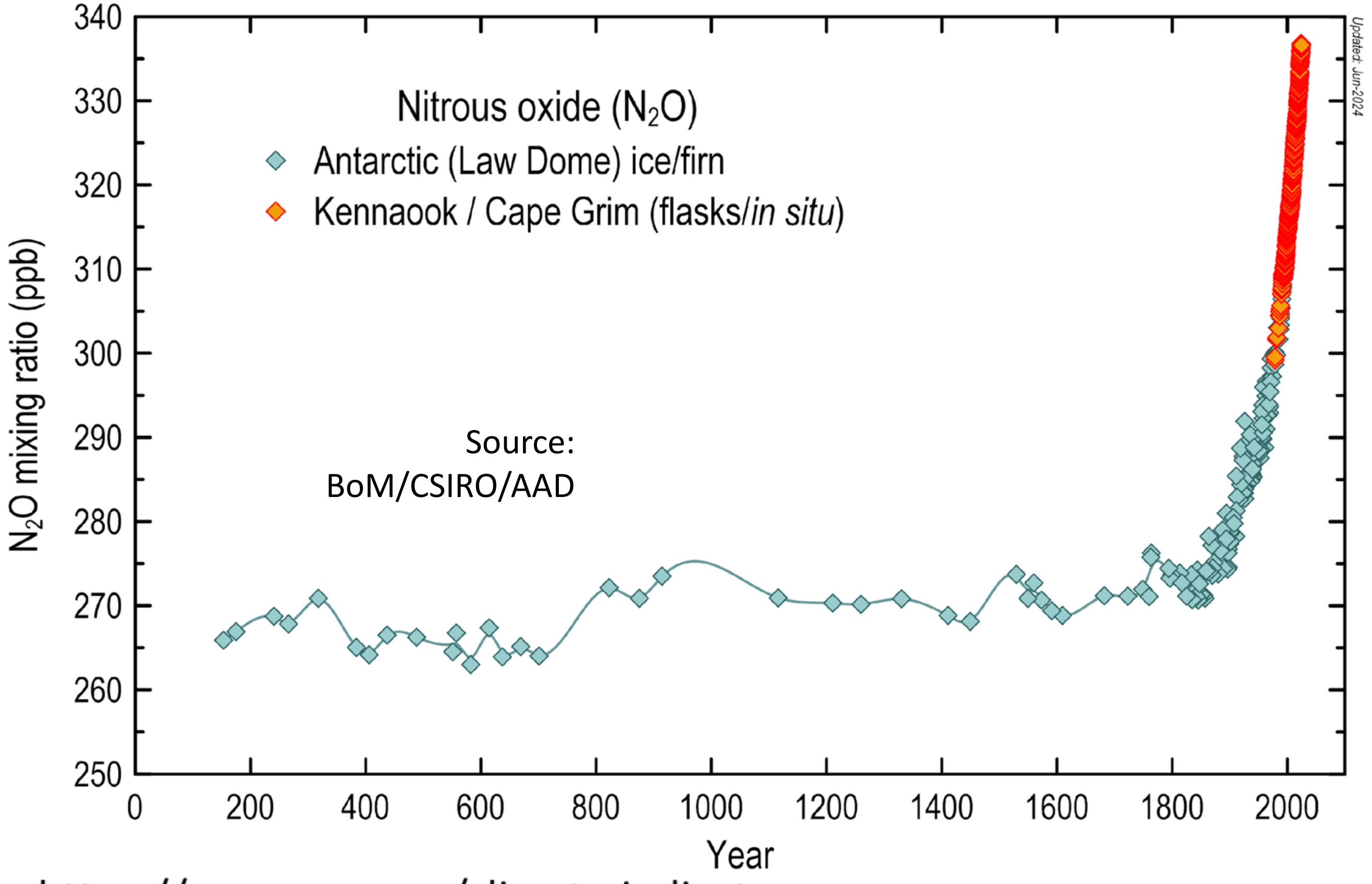
Human influence on the global N cycle



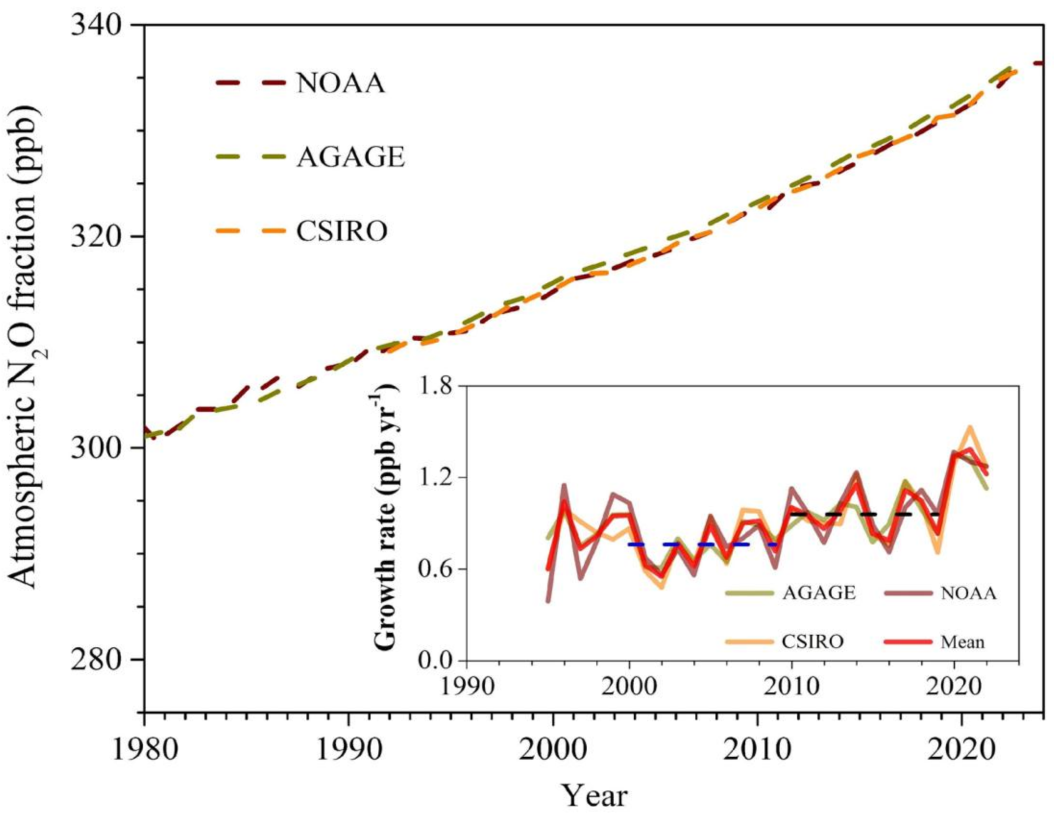
- **Haber-Bosch process:** nitrogen-based synthetic fertilizers
  - Uses natural gas
  - GHG emissions
- Fossil fuels combustion
- Wastewater discharge
- Agricultural runoff **(> 65 Tg N/Yr)**

Doubling of the "N" compounds in the past 100 years

Atmospheric N_2O Concentrations



https://www.epa.gov/climate-indicators



The global N_2O *concentration* has increased by about 25%, from 270 parts per billion (ppb) in 1750 to 336 ppb in 2022

The mean growth rate for 2010-2019 was 0.96 ppb yr^{-1}.

Source: Global Carbon project

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The world's forgotten greenhouse gas

(Image credit: Getty Images)

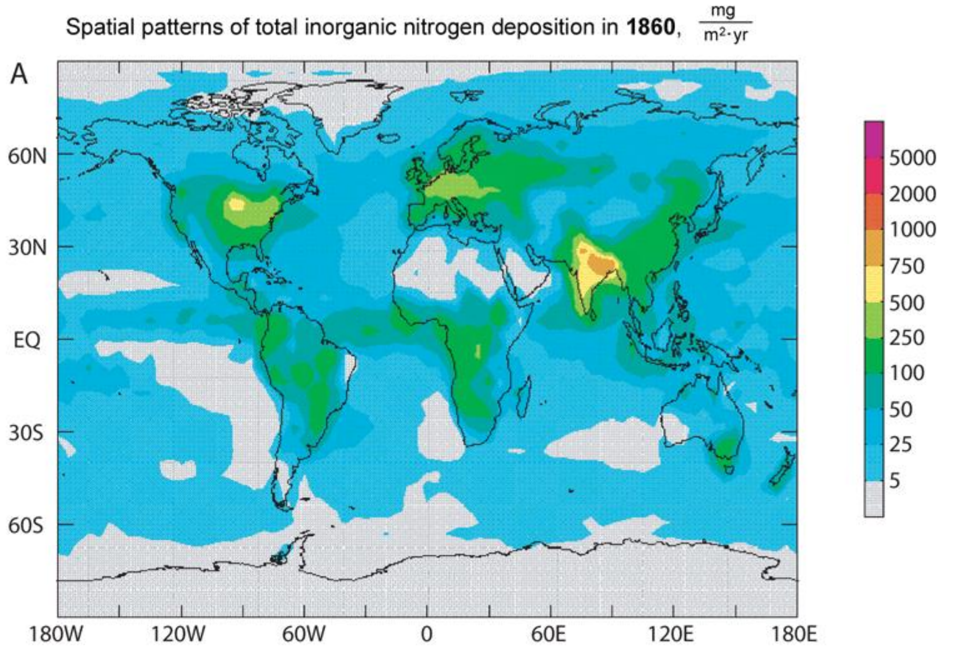
https://www.bbc.com/future/article/20210603-nitrous-oxide-the-worlds-forgotten-greenhouse-gas

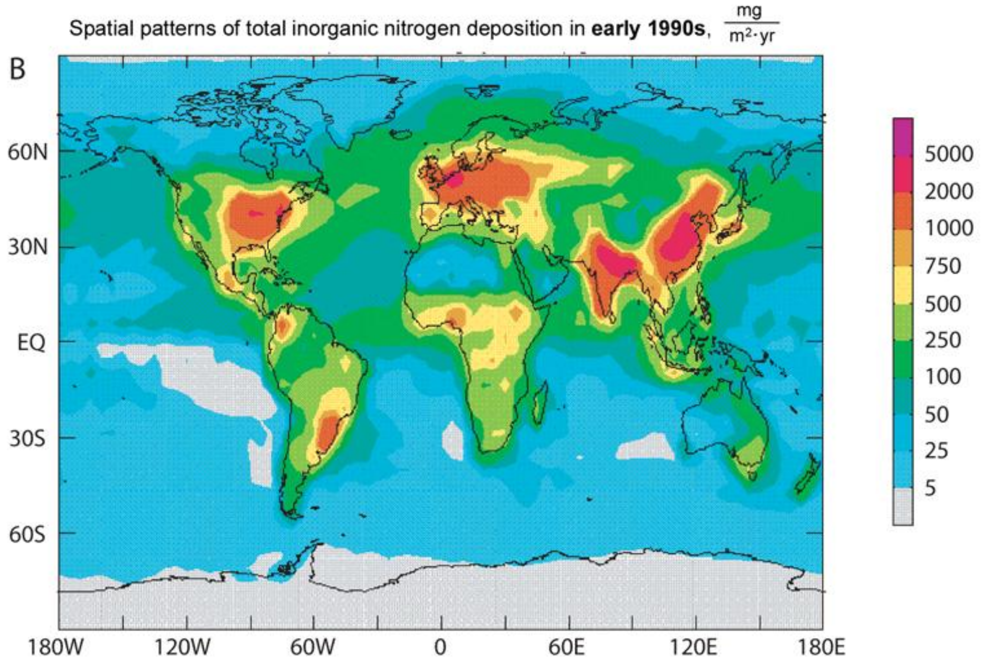
By Ula Chrobak 4th June 2021

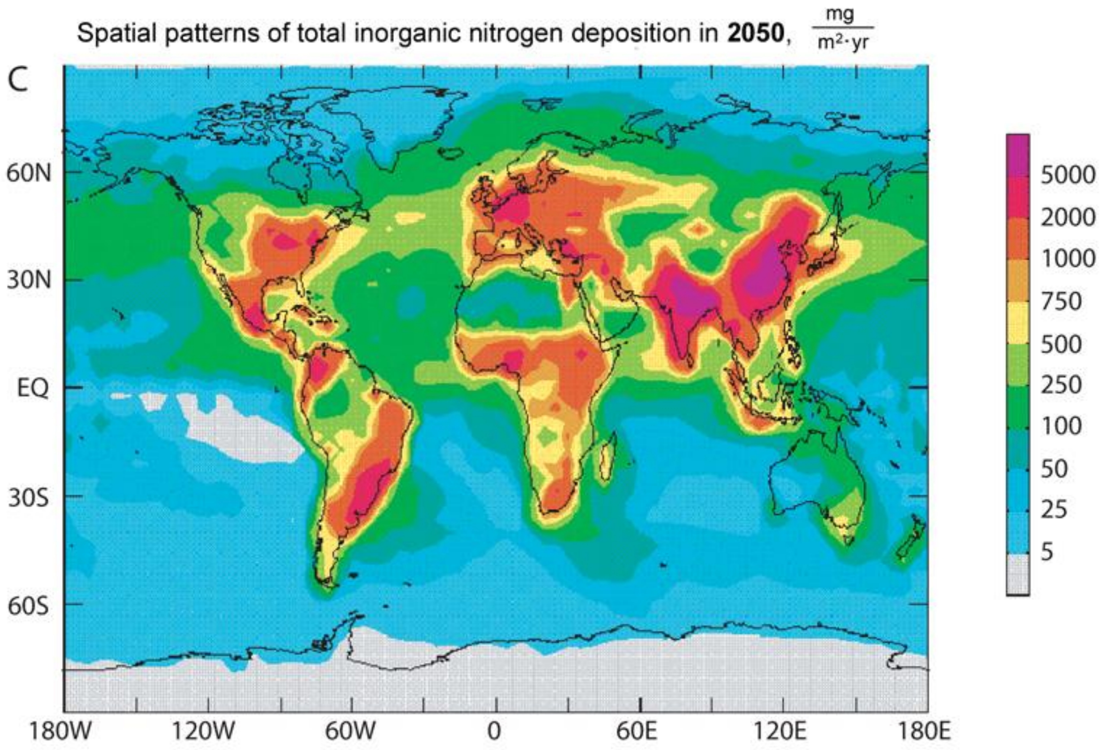
From Knowable Magazine

Emissions of the greenhouse gas commonly known as laughing gas are soaring. Can we cut emissions from its greatest anthropogenic source?

Global N (im)balance – high input (e.g., nitrates), limited recycling







By 2050, biologically reactive nitrogen will be widespread worldwide.

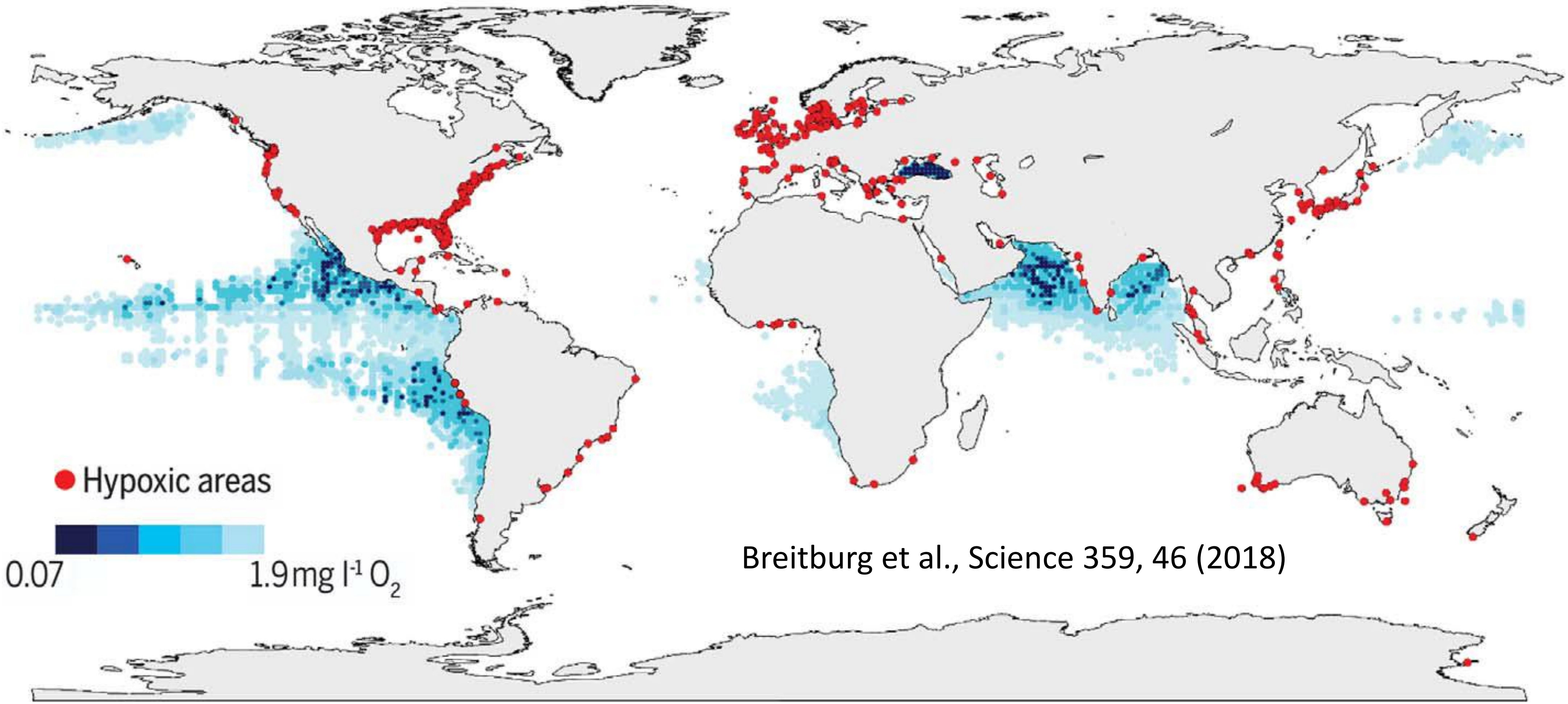
Credit: F. J. Dentener

Impacts of reactive N (e.g., nitrate, ammonium, etc.) to water and soil ecosystems

Dal lake, with and w/o eutrophication

Pic credit: Faisal Khan

Low and declining oxygen levels in the open ocean and coastal waters



Breitburg et al., Science 359, 46 (2018)

Rapid increase in the aquatic dead zones across the world

Consequences of increased reactive N in environment

- Impacts on

  soils (e.g., acidification)

  atmosphere (e.g., Ozone depletion, greenhouse gas balance; N_2O 270 ppb in 1750 to 336 ppb in 2022 (25%↑))

  water sources (e.g., groundwater nitrate contamination – up to 200 mg/L in Malwa region, Punjab) and

  freshwater/seawater (e.g., eutrophication and dead zones)

- Affect climate, ecosystems, and human health